Curriculum Vitae

Personal Information

Name: WU, Jingxian
Gender: Male
Tel: (573) 884-0097
Date of Birth: November 25, 1975
Major: Electrical and Computer Engineering
E-Mail: jwu@ee.missouri.edu

Education

2001 ~ Present **University of Missouri – Columbia** **MO, USA**
Candidate of Ph.D.
- Grade Average: 4.0/4.0

1998 ~ 2001 **Tsinghua University** **Beijing, P. R. China**
Master of Science, Jan. 2001
- Ranked top 3 out of 72
- Grade Average: 88.6/100

1994 ~ 1998 **Beijing University of Aero. & Astro.** **Beijing, P. R. China**
Bachelor of Engineering, July 1998
- Ranked top 2 out of 112
- Grade Average: 89.0/100

Research Experiences

2001 ~ Present **The Communication Lab** **MU**
Research Assistant
- Discrete-time channel modeling for MIMO wireless communication systems
- Multi-user channel estimation for broadband CDMA communication systems
- Channel estimation for wireless communication systems with pilot symbols
- Frequency-domain equalization with space-time trellis coding
- Simulation model development for systems with non-isotropic scattering fading channels

1999 ~ 2001 **The Broadband Access Network Lab** **Tsinghua University**
Research Assistant
- Build the network layer simulation modules for hybrid fiber coaxial (HFC) network with C++ and the Network Simulator *ns*-2
- Proposed a set of MAC layer resource allocation schemes of HFC access network
- Developed the embedded operational software for Cable Modem, on the platform of PSOS
- Designed and implemented the hardware architecture of Cable Modem
- Researched on the return path noise of HFC network

1998 ~ 1999 **The Tsinghua–Motorola Personal Communication Lab Tsinghua University**

Research Assistant

- Researched on the software architecture of embedded Real Time Operating System (RTOS)
- Designed a dictionary engine for Personal Digital Assistant (PDA). This engine includes fast search algorithm and compression algorithm, and can be implemented on various embedded platforms
- A developing member of Motorola's PDA product MobilePAD828™
- A developing member of Motorola's mobile phone product A6188™

1997 ~ 1998 **The Synthetic Aperture Radar Research Lab BUAA**

Research Assistant as an undergraduate student

- Independently designed and developed the software of VisualSAR, a visualized and integrated SAR image analysis system on the UNIX platform

1997 **The Robot Research Institute BUAA**

Research Assistant as an undergraduate student

- Designed and implemented an interface circuit for a high-building-window-cleaning Robot. This interface circuit interacts with both the micro-computer and the mechanical control part of the

Professional Activities

- Reviewer for IEEE Journal on Selected Areas of Communications
- Reviewer for IEEE Transactions on Wireless Communications
- Reviewer for IEEE Transactions on Transaction on Vehicular Technology
- Reviewer for IEEE Transactions on Circuits and Systems - I
- Reviewer for International Journal on Wireless Information Networks
- Internship engineer in Mitsubishi Electric Research Lab (MERL) (2003)
- Internship engineer in the Beijing I-Star Company (1999)

Academic Honors

- Research Assistantship (2001, 2002, MU)
- Guanghua Fellowship (1999, Tsinghua University**)**
- Internship Scholarship, First Prize (1999, Tsinghua University)
- Privilege to be enrolled in the master program of Tsinghua University with entrance examination waived (1998, Tsinghua University)
- The People's Scholarship, First Prize (1995, 1996, 1997, BUAA)
- Second Awards of the Advanced Mathematical Competition (1996, BUAA)
- Lizheng Fellowship for the top 1 freshman (1994, BUAA)

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Skills

- Proficient with computer programming in C/C++ on both Windows and UNIX platform
- Communication systems simulation with Matlab
- Network layer simulation for communication networks
- Familiar with the script langrage Tcl/Tk
- Software design and hardware implementation of communication systems
- Familiar with PCB design tools, such as PowerPCB, Protel, ORCAD
- Linux system administration

Papers

Journal Papers

- Jingxian Wu, Chengshan Xiao, and Jan C. Olivier, "Time-Varying and frequency-selective channel estimation with unequally spaced pilot symbols", submitted to *International J. Wireless Info. Networks*, 19 pages, 2003.
- Chengshan Xiao, Jingxian Wu, Sang-Yick Leong, Yahong R. Zheng and Khaled B. Letaief, "A Discrete-time model for spatio-temporally correlated MIMO WSSUS multipath channels", accepted for publication by *IEEE Trans. Wireless Commun.*, 28 pages, 2003.
- Jingxian Wu and Chengshan Xiao, "Multiuser channel estimation for CDMA systems over Doubly Selective fading channels", submitted to *IEEE Trans. Wireless Commun.*, 26 pages, 2003.
- Jingxian Wu, and Chengshan Xiao, "Error Performance of Optimum Coherent Diversity Receivers with Channel Estimation Error in Rayleigh and Ricean Fading Channels", ready for submission, 24 pages, 2003.

Conference Papers

- Jingxian Wu, Chengshan Xiao, and Jan C. Olivier, "Time-Varying and frequency-selective channel estimation with unequally spaced pilot symbols", in *Porc. IEEE Intern. Conf. Acoustic Speech & Signal Processing ICASSP'03*, Apr. 2003.
- Chengshan Xiao, Jingxian Wu, and Sang-Yick Leong, "A Discrete-time model for spatio-temporally correlated MIMO WSSUS multipath channels", in *Proc. IEEE Wireless Commun. Network Conf. WCNC'03*, Mar. 2003.
- Jingxian Wu, Chengshan Xiao, and Khaled B. Letaief, "MMSE-Based Channel Estimation for CDMA Systems over Frequency-Selective Fading Channels", to appear in *Proc. IEEE Vehicular Techno. Conf. VTC'03 Fall*, 5 pages, 2003.
- Sang-Yick Leong, Jingxian Wu, Jan C. Olivier, and Chengshan Xiao, "Fast Time-varying Dispersive Channel Estimation and Equalization for 8-PSK Cellular System", to appear in *Proc. IEEE Vehicular Techno. Conf. VTC'03 Fall*, 5 pages, 2003.
- Jingxian Wu, Mingquan Lu, and Zhenming Feng, "Extended ns for HFC MAC Layer Simulation", in *Proc. Intern. Conf. Telecommun. ICT'02*, Jun. 2002.

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